

Microeconomics Problem

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1. The total utility function of a consumer is

$$TU = 12X - X^2.$$

- (1) Find the marginal utility function.
 - (2) What is marginal utility when $X = 3$?
 - (3) At what level of consumption is total utility maximized?
2. The price of a good rises from 10 to 12, and quantity demanded falls from 50 to 45. Use the percentage-change method to calculate the price elasticity of demand.
3. A consumer's income rises by 25%, and the demand for a good rises by 10%. Find the income elasticity of demand and classify the good.
4. The price of apples rises by 20%, and the demand for oranges rises by 8%. Find the cross-price elasticity of demand and determine whether the goods are substitutes or complements.
5. The demand function is

$$Q_d = 30 - 2P,$$

and the supply function is

$$Q_s = 6 + P.$$

- (1) Find the equilibrium price and quantity.
 - (2) Find the price elasticity of demand and the price elasticity of supply at equilibrium.
6. The demand function is

$$Q_d = 20 - P,$$

and the supply function is

$$Q_s = 4 + P.$$

The government imposes a tax of 2 per unit on sellers. Find the price paid by buyers, the price received by sellers, and the quantity traded after the tax.

7. A firm's production function is

$$Q = 6L - L^2.$$

- (1) Find the average product of labor.
- (2) Find the marginal product of labor.
- (3) At what level of labor input is total output maximized?

8. Suppose the production function is

$$Q = K^{1/2}L^{1/2}.$$

Determine whether the production function exhibits constant returns to scale.

9. A firm's production function is

$$Q = 10L - L^2.$$

The output price is 4, and the wage rate is 16. Capital is fixed. Find the profit-maximizing labor input.

10. The short-run total cost function is

$$STC = Q^2 + 10Q + 36.$$

Find:

- (1) fixed cost,
- (2) variable cost,
- (3) average cost,
- (4) marginal cost.

11. A firm's marginal cost function is

$$MC = 4Q + 2,$$

and fixed cost is 10. Find the total cost function.

12. A perfectly competitive firm's total cost function is

$$TC = Q^2 + 4Q + 25,$$

and the market price is

$$P = 16.$$

Find the profit-maximizing output and profit.

13. A perfectly competitive firm's short-run total cost function is

$$TC = Q^2 + 8Q + 16.$$

Find the shutdown price.

14. A monopolist faces the demand function

$$P = 20 - Q,$$

and the total cost function

$$TC = 4Q + 10.$$

Find the profit-maximizing output, price, and profit.

15. In the previous question, suppose the market becomes perfectly competitive and marginal cost remains 4. Find the competitive output and the deadweight loss caused by monopoly.

16. Two firms produce a homogeneous product. The inverse market demand is

$$P = 30 - Q_1 - Q_2,$$

and both firms have constant marginal cost equal to 6. Find the Cournot equilibrium outputs, total output, market price, and each firm's profit.

17. Two Cournot firms face market demand

$$P = 24 - Q_1 - Q_2,$$

with marginal costs

$$MC_1 = 2, \quad MC_2 = 4.$$

Find the Cournot equilibrium outputs.

18. In the previous question, find the market price and each firm's profit.